Andrew Janigian

2/10/2026 - 2/24/2026

HALEU Supply Crisis: Summary and Analysis

Small Modular Reactors depend on High Assay Low Enriched Uranium (HALEU), which is uranium enriched to between 5 and 20 percent uranium 235. Natural uranium contains about 0.7 percent uranium 235, while the remainder is uranium 238. Conventional light water reactors operate with fuel enriched to between 3 and 5 percent uranium 235. The increase above 5 percent alters core physics, fuel cycle length, and reactor configuration in ways that are central to the operation of many Small Modular Reactor (SMR) designs.

Without getting too technical, Uranium 235 sustains the chain reaction that produces heat in a nuclear reactor. Increasing its concentration improves neutron economy and thereby allows a given mass of fuel to support longer operating cycles. The benefits of this are numerous, but for the purposes of this paper I will discuss specifically how this pertains to SMRs. Many Small Modular Reactor concepts are designed for extended operation without refueling and for compact cores that rely on higher enrichment to maintain criticality over time. Core dimensions, fuel rod spacing, control rod insertion limits, and burnup targets are calculated using a defined enrichment level. If a reactor designed for 15 percent enrichment were loaded with fuel enriched to 4.5 percent, the reduction in reactivity would alter power output, cycle length, and shutdown

margins. Correcting for this change would require redesign of the core and new licensing analysis. Substitution of conventional fuel is therefore not a practical solution.

Commercial enrichment currently relies on uranium hexafluoride (UF_6), a compound formed by reacting uranium oxide with fluorine. Uranium hexafluoride is used because it transitions to a gas at a moderate temperature (~56 °C or ~134 °F), which permits separation of isotopes in a gaseous state. In gas centrifuge enrichment, uranium hexafluoride is fed into rapidly rotating cylinders. The slight mass difference between uranium 235 and uranium 238 causes molecules containing uranium 238 to move toward the outer wall, while molecules containing uranium 235 remain closer to the center. The separation achieved in a single centrifuge is limited because the mass difference between isotopes is small. This is why the enrichment process is commonly run multiple times, as to best isolate the desired product.

Industrial production requires cascades composed of many centrifuges connected in series and parallel. Gas flows from one stage to the next, with each stage increasing the concentration of uranium 235 by a small increment. The number of stages and the total separative work capacity determine the final enrichment level. Enrichment to 15 percent requires more separative work units per kilogram of product than enrichment to 4 percent, due to returns diminishing during the final stages of enrichment. Facilities designed to supply fuel at 3 to 5 percent are configured with cascade arrangements optimized for that range and cannot produce material above 5 percent without reconfiguration and sufficient installed capacity. To put it another way, it is unlikely that domestic fuel synthesis could occur, as the cost of reconfiguration is simply too high.

Further, handling uranium hexafluoride presents significant chemical and environmental risks. When uranium hexafluoride reacts with water vapor, it forms hydrogen fluoride (HF) and uranyl fluoride (UO_2F_2). Hydrogen fluoride is toxic and corrosive, and exposure can damage tissue and respiratory systems. For this reason, enrichment facilities must prevent moisture contamination, maintain sealed systems, and use materials compatible with fluorine chemistry. Leak detection, pressure control, and containment systems are required to reduce the risk of release. This further narrows down the potential for domestic synthesis to occur, both due to potential environmental contamination from uranium compounds (in the event of accidental discharge) and because our enrichment facilities may not be capable of quarantining to the degree that is required.

As of January 2026, the United States has no commercial scale domestic supply of HALEU; global supply has been concentrated in Russia through Rosatom. Sanctions and trade restrictions have limited access to this source for United States companies, creating a supply constraint that affects all reactor developers that depend on uranium enrichment above 5 percent. To remedy this, the Department of Energy has initiated programs to support domestic enrichment capability, including funding for demonstration scale production and initial stockpiles. These efforts have not yet resulted in sustained commercial output sufficient to support multiple reactor deployments though. Centrus Energy is developing enrichment capacity in Kentucky with the objective of producing High Assay Low Enriched Uranium at commercial scale. Public disclosures indicate that broader commercial operation is projected for 2028 at the earliest, with initial production volumes before that date remaining limited.

This timeline creates a gap between potential reactor licensing and practical deployment. If advanced reactor licensing frameworks are finalized before 2028, developers such as NuScale Power and TerraPower may obtain approval to construct and operate their designs. However, approval does not guarantee the stable supply of fuel. Initial core loading requires a defined quantity of enriched uranium produced and fabricated in advance of startup. Without sufficient enrichment capacity, reactors cannot begin operation even if construction is complete. This is the crux of the supply crisis, and why fission energy production has slowed since the onset of the Ukraine war.

```
Uranium Mining
      |
      ▼
Conversion to UF6
      |
      ▼
Enrichment Capacity
      |
      ├  Rosatom (Russia)
      |       |
      |       └── Sanctions restrict US access
      |
      └── Centrus Energy (Kentucky)
              |
              ├── Demonstration scale production
              ├  Scale up period 2026 to 2028
              └── Commercial scale earliest 2028
      |
      ▼
Fuel Fabrication for SMRs
      |
      ▼
SMR Core Loading and Operation
```

Figure 1: Rough Supply Chain Flowchart (visualized using Python)